

D5: Osterwalder-Seiler Rate/Summability Certificate (H2a)

Machine Certificate v1.6 | David J. Fox | June 9, 2026

Opera Numerorum -- YM Tower -- Wall256 Leaf D5

1. Claim

D5 (concrete geometric activity): Given $w_1 = \text{chi_prime}(\beta_0)$ satisfying $0 < w_1 < 1/7$ (certified by D4), the concrete polymer activity $a_n = w_1^n$ satisfies TruncatedActivityBound: (1) nonnegativity: $w_1^n \geq 0$ for all n ; (2) rate $I = -\log(w_1) > \log(7)$ (strict); (3) per-size bound: $w_1^n \leq \exp(-I)^n$ (equality); (4) entropy-weighted geometric series $\sum_n 7^n \cdot \exp(-I)^n$ converges.

Open condition (E4): The actual SU(3) lattice polymer activity is a Ursell/truncated cluster integral, NOT simply w_1^n . The Osterwalder-Seiler 1978 Thm 2.1 product bound $|\text{z_T}(\gamma)| \leq \prod_{p \in \gamma} |\exp(-\beta S_p)|$ is absent from Mathlib v4.12.0. H2b stays OPEN / OUT_OF_TOWER.

2. Certified Numerical Values

```
beta_0      = 2.079416880124
w1(beta_0)   = 0.14285714285633053    [chi_prime.py, D4 certified]
1/7          = 0.14285714285714285
D4 gap (1/7-w1) = 8.123224315426114e-13 > 0 [D4 HOLDS]

I = -log(w1)  = 1.9459101490609996
log(7)        = 1.9459101490553132
I - log(7)    = 5.686340287525127e-12 > 0 [RATE HOLDS]

z = 7 * w1    = 0.9999999999943137    < 1 [GEOM HOLDS]
sum z^n = 1/(1-z) < inf                [CONVERGENT]
```

3. Mathematical Argument

Given $0 < w_1 < 1/7$:

1. NONNEGATIVITY: $w_1 > 0 \Rightarrow w_1^n \geq 0$. HOLDS.

2. RATE: Set $I := -\log(w_1)$.
 $w_1 < 1/7$
 $\Rightarrow \log(w_1) < \log(1/7) = -\log(7)$
 $\Rightarrow -\log(w_1) > \log(7)$
 $\Rightarrow I > \log(7)$. HOLDS.

3. PER-SIZE BOUND:
 $w_1^n = \exp(n \cdot \log(w_1)) = \exp(-I \cdot n) = \exp(-I)^n$. HOLDS (equality).

4. SUMMABILITY:
 $\sum_{n \geq 0} 7^n \cdot \exp(-I)^n$
 $= \sum_{n \geq 0} (7 \cdot w_1)^n$
 $= 1 / (1 - 7w_1)$ [geometric, since $0 < 7w_1 < 1$]
 $< \text{inf}$. HOLDS.

4. Lean Machine-Check (Wall256_OS.lean)

The following theorems in **lean-proof-towers/Towers/YM/Wall256_OS.lean** formalize the argument above in Lean 4. Axiom footprint: classical trio only. No sorry. No extra axiom.

```
-- os_geometric_activity_bound (line 155)
theorem os_geometric_activity_bound (w1 : R) (hw1_pos : 0 < w1) (hw1 : w1 < 1/7) :
  Wall256Note.TruncatedActivityBound (fun n => w1 ^ n)
```

```
-- hOS_concrete (line 179)
theorem hOS_concrete (w1 : R) (hw1_pos : 0 < w1) :
  w1 < 1/7 ->
  Wall256Note.TruncatedActivityBound (fun n => w1 ^ n)

-- Axiom footprint: {propext, Classical.choice, Quot.sound}
-- No sorry. No axiom. Classical trio preserved.
```

5. Open Conditions (explicitly stated)

H2b (E4) -- OPEN: OS product bound for actual SU(3) polymer activity. The actual cluster weight $z_T(\gamma)$ for a connected polymer γ of n plaquettes is a Ursell/truncated cluster integral over $SU(3)^{\{\text{links}\}}$. Osterwalder-Seiler 1978 Thm 2.1 asserts $|z_T(\gamma)| \leq \prod_{p \in \gamma} w_1$, making z_T bounded by w_1^n . This step is absent from Mathlib v4.12.0. Until formalized, D5 for the actual polymer model stays OPEN.

E1 -- OPEN: SU(3) Haar integral link to w_1 . The abstract w_1 in the Lean scaffold is not yet connected to the concrete SU(3) integral $\chi_{\text{prime}}(\beta)$. The β_0 threshold is CERT_Arb certified out-of-tower (mpmath.iv, $N=36$), not a Lean term.

Surface #1 / YM mass gap -- OPEN. D5 (even fully proved) closes only the lattice activity bound. The continuum limit and $m > 0$ (Clay problem) stay OPEN.

6. Cryptographic Binding

Item		SHA-256 Digest
Source	certificates/d5_cert.py	d3edea8718750db8d38dfbc64f8277472f483557b7a9c0ad603a43c183b022e5
Stdout	d5.out	09341d4bf50b1f8a3261322059e2ec959100bca8f7f4d92a097f66ff26f18d02
Lean	wall256_OS.lean	c6936035c4ab391978274eb1fb000939e727a06e3889d9d300705ac02db2362d

7. D5 Status Summary

Component	Status
D4: $w_1(\beta_0) < 1/7$	CERTIFIED (chi_prime.py)
H2a: $\text{rate } I = -\log(w_1) > \log(7)$	MACHINE-CHECKED (Wall256_OS.lean)
H2a: $a_n = w_1^n$ satisfies TruncActBound	MACHINE-CHECKED (Wall256_OS.lean)
H2b: OS product bound (actual polymer)	OPEN -- E4, absent Mathlib v4.12.0
E1: SU(3) Haar link to w_1	OPEN -- out-of-tower (CERT_Arb)
Surface #1 / YM mass gap / Clay	OPEN -- no claim

Reproduce: `PYTHONPATH=certificates python3 certificates/d5_cert.py | sha256sum`

Depends on D4: chi_prime_source (SHA 5ea6404488c2aff5...) -- Wall256_OS.lean (SHA c6936035c4ab3919...) -- Opera Numerorum / Battle Plan v1.6 -- David J. Fox